

Practice 1

2520030575

M.LIKHITHA

Create a database & use:

```
1 CREATE DATABASE IF NOT EXISTS bookflow2_db;  
2 USE bookflow_db;
```

Output

Action Output

#	Time	Action
✓ 1	09:08:46	SHOW DATABASES
✓ 2	09:09:23	CREATE DATABASE IF NOT EXISTS bookflow2_db
✓ 3	09:09:40	USE bookflow_db

Create a physician table:

```
3 CREATE TABLE physician (  
4     employeeid INT PRIMARY KEY,  
5     name VARCHAR(100),  
6     position VARCHAR(100),  
7     ssn VARCHAR(20)  
8 );
```

Output

Action Output

#	Time	Action	Message
✓ 1	14:56:17	CREATE DATABASE IF NOT EXISTS bookflow5_db	1 row
✓ 2	14:56:29	USE bookflow5_db	0 row
✓ 3	14:59:46	CREATE TABLE physician (employeeid INT PRIMARY KEY, name V...	0 row

Insert data into physician:

```

9 • INSERT INTO physician VALUES
10 (1,'Likhitha','Staff Internist','11111111'),
11 (2,'Kavya','Attending Physician','22222222'),
12 (3,'Divya','Surgical Attending Physician','33333333'),
13 (4,'Akhila','Senior Attending Physician','44444444'),
14 (5,'Sushma','Head Chief of Medicine','55555555'),
15 (6,'Vaishnavi','Surgical Attending Physician','66666666'),
16 (7,'Navya','Surgical Attending Physician','77777777'),
17 (8,'Sai','MD Resident','88888888'),
18 (9,'Shiva','Attending Psychiatrist','99999999');
19
20

```

Output				
Action Output				
	#	Time	Action	Message
✓	14	15:03:56	DROP DATABASE bookflow5_db	1 row(s) affected
✓	15	15:04:01	DROP DATABASE bookflow6_db	0 row(s) affected
✓	16	15:04:07	CREATE DATABASE IF NOT EXISTS bookflow6_db	1 row(s) affected
✗	17	15:04:09	USE bookflow5_db	Error Code: 1049. Ur
✓	18	15:04:14	USE bookflow6_db	0 row(s) affected
✓	19	15:04:17	CREATE TABLE physician (employeeid INT PRIMARY KEY, nam...	0 row(s) affected
✓	20	15:04:19	INSERT INTO physician VALUES (1,'Likhitha','Staff Internist','11111111...	9 row(s) affected Re

Select:

```

19 • SELECT * FROM physician;
20
21

```

Result Grid				
Filter Rows:				
Edit:    				
	employeeid	name	position	ssn
▶	1	Likhitha	Staff Internist	111111111
	2	Kavya	Attending Physician	222222222
	3	Divya	Surgical Attending Physician	333333333
	4	Akhila	Senior Attending Physician	444444444
	5	Sushma	Head Chief of Medicine	555555555
	6	Vaishnavi	Surgical Attending Physician	666666666
	7	Navya	Surgical Attending Physician	777777777
	8	Sai	MD Resident	888888888
	9	Shiva	Attending Psychiatrist	999999999
▲	NULL	NULL	NULL	NULL

Create department table:

```

20 • CREATE TABLE department (
21     departmentid INT PRIMARY KEY,
22     name VARCHAR(100),
23     head INT
24 );
25

```

Output

Action Output

#	Time	Action
✓ 20	15:04:19	INSERT INTO physician VALUES (1,'Likhitha','Staff Internist','11111111...
✓ 21	15:04:57	SELECT * FROM physician LIMIT 0, 1000
✓ 22	15:05:39	CREATE TABLE department (departmentid INT PRIMARY KEY, n...

Insert data into department:

```

25 • INSERT INTO department VALUES
26     (1,'General Medicine',4),
27     (2,'Surgery',7),
28     (3,'Psychiatry',9);
29 • SELECT * FROM department

```

Result Grid Filter Rows: Ed

	departmentid	name	head
▶	1	General Medicine	4
	2	Surgery	7
	3	Psychiatry	9
*	NULL	NULL	NULL

Inner join

```

30 • SELECT d.name AS Department,
31         p.name AS Head_Physician
32 FROM department d
33 JOIN physician p
34 ON d.head = p.employeeid;

```

Result Grid Filter Rows: Ed

	Department	Head_Physician
▶	General Medicine	Akhila
	Surgery	Navya
	Psychiatry	Shiva

Where clause:

```

35 • SELECT *
36 FROM physician
37 WHERE position='Surgical Attending Physician';

```

Result Grid Filter Rows: Edit

	employeeid	name	position	ssn
▶	3	Divya	Surgical Attending Physician	333333333
	6	Vaishnavi	Surgical Attending Physician	666666666
	7	Navya	Surgical Attending Physician	777777777
*	NULL	NULL	NULL	NULL

Where + Like:

```
38 • SELECT *
39 FROM physician
40 WHERE name LIKE 'Likhitha%';
```

Result Grid | Filter Rows: | Edit

	employeeid	name	position	ssn
▶	1	Likhitha	Staff Internist	111111111
•	NULL	NULL	NULL	NULL

Count():

```
41 • SELECT COUNT(*) AS Total_Physicians
42 FROM physician;
```

Result Grid | Filter Rows: | Export

	Total_Physicians
▶	9

Count Distinct:

```
43 • SELECT COUNT(DISTINCT position)
44 FROM physician;
```

Result Grid | Filter Rows: | Export

	COUNT(DISTINCT position)
▶	7

Group by:

```
45 • SELECT position,
46 COUNT(*) AS Total_Employees
47 FROM physician
48 GROUP BY position;
```

Result Grid | Filter Rows: | Export

	position	Total_Employees
▶	Staff Internist	1
	Attending Physician	1
	Surgical Attending Physician	3
	Senior Attending Physician	1
	Head Chief of Medicine	1
	MD Resident	1
	Attending Psychiatrist	1

Group by + having:

```

49 • SELECT position,
50       COUNT(*) AS Total_Employees
51 FROM physician
52 GROUP BY position
53 HAVING COUNT(*) > 1;

```

Result Grid			Filter Rows:	Export
	position	Total_Employees		
▶	Surgical Attending Physician	3		

Order by:

```

54 • SELECT *
55 FROM physician
56 ORDER BY name ASC;

```

Result Grid

Filter Rows:

Edit:

	employeeid	name	position	ssn
▶	4	Akhila	Senior Attending Physician	444444444
	3	Divya	Surgical Attending Physician	333333333
	2	Kavya	Attending Physician	222222222
	1	Likhitha	Staff Internist	111111111
	7	Navya	Surgical Attending Physician	777777777
	8	Sai	MD Resident	888888888
	9	Shiva	Attending Psychiatrist	999999999
	5	Sushma	Head Chief of Medicine	555555555
	6	Vaishnavi	Surgical Attending Physician	666666666
*	NULL	NULL	NULL	NULL

Order by desc:

```

57 • SELECT *
58 FROM physician
59 ORDER BY employeeid DESC;

```

Result Grid

Filter Rows:

Edit:

	employeeid	name	position	ssn
▶	9	Shiva	Attending Psychiatrist	999999999
	8	Sai	MD Resident	888888888
	7	Navya	Surgical Attending Physician	777777777
	6	Vaishnavi	Surgical Attending Physician	666666666
	5	Sushma	Head Chief of Medicine	555555555
	4	Akhila	Senior Attending Physician	444444444
	3	Divya	Surgical Attending Physician	333333333
	2	Kavya	Attending Physician	222222222
	1	Likhitha	Staff Internist	111111111
*	NULL	NULL	NULL	NULL

Where + in:

```

60 • SELECT *
61 FROM physician
62 WHERE employeeid IN
63 (
64     SELECT head
65     FROM department
66 );

```

Result Grid

	employeeid	name	position	ssn
▶	4	Akhila	Senior Attending Physician	444444444
	7	Navya	Surgical Attending Physician	777777777
	9	Shiva	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

Subquery:

```

67 • SELECT *
68 FROM physician
69 WHERE employeeid NOT IN
70 (
71     SELECT head
72     FROM department
73 );

```

Result Grid

	employeeid	name	position	ssn
▶	1	Likhitha	Staff Internist	111111111
	2	Kavya	Attending Physician	222222222
	3	Divya	Surgical Attending Physician	333333333
	5	Sushma	Head Chief of Medicine	555555555
	6	Vaishnavi	Surgical Attending Physician	666666666
	8	Sai	MD Resident	888888888
*	NULL	NULL	NULL	NULL

Join + where

```

74 • SELECT p.name
75 FROM physician p
76 JOIN department d
77 ON p.employeeid=d.head
78 WHERE d.name='Surgery';

```

Result Grid

	name
▶	Navya

Join + order by:

```

79 • SELECT d.name,
80         p.name
81     FROM department d
82    JOIN physician p
83   ON d.head=p.employeeid
84   ORDER BY d.name;

```

Result Grid | Filter Rows:

	name	name
▶	General Medicine	Akhila
	Psychiatry	Shiva
	Surgery	Navya

Count + having

```

85 • SELECT position,
86         COUNT(*) AS Total
87     FROM physician
88    GROUP BY position
89   HAVING COUNT(*) >= 2;

```

Result Grid | Filter Rows:

	position	Total
▶	Surgical Attending Physician	3

Exists:

```

90 • SELECT *
91     FROM department d
92    WHERE EXISTS
93    (
94        SELECT 1
95        FROM physician p
96       WHERE p.employeeid=d.head
97    );

```

Result Grid | Filter Rows:

	departmentid	name	head
▶	1	General Medicine	4
	2	Surgery	7
	3	Psychiatry	9
*	NULL	NULL	NULL

Max():

```
98 • SELECT MAX(employeeid)
99 FROM physician;
```

Result Grid   Filter Rows:

	MAX(employeeid)
▶	9

Min():

```
100 • SELECT MIN(employeeid)
101 FROM physician;
```

Result Grid   Filter Rows:

	MIN(employeeid)
▶	1